

LEARNING INTENTIONS

Students will be able to:

- Construct appropriate graphs using given information.
- Interpret the relationship between variables in a graph.
- Use a graph as a source of data.

Graphs

Graphing is a very good way to represent data that you collect during a survey or experiment. It allows you to quickly notice a trend or relationship between the variables you are investigating.

Different Types of graphs can be used to represent data:

1. Line Graphs
2. Bar Charts
3. Pie Charts

Independent Variable: the variable that you change in an investigation.

This data goes on the X-axis

Dependent Variable: Variable that you measure in an investigation.

This data goes on the Y-axis

Steps in drawing a graph:

1. Always use graph paper.
2. Draw the X and Y axis.
3. Label the axes with the correct variable and units.
4. Decide the scale for each axis. Remember there has to be equal jumps on each axis.
5. Plot the data. Make sure you clearly mark the point.
6. Join the points with a line or curve.
7. Give your graph a title.

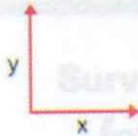
Tools:

- Pencil
- Graph paper
- Ruler

S P L A T
C O A X I
A J B I T
L N E S L E
E T L E
S S

Draw a large
clear graph!

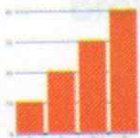
GRAPHING INFORMATION



The **axis** is made up of reference lines used to create your graph. There is an x-axis (horizontal) and a y-axis (vertical).



The **title** tells viewers what your graph is about. Make sure you choose an appropriate title that tells viewers what your graph is about.



Data is the information that you are showing on your graph. Data is usually the last thing that you should put on your graph.

Scale

The **scale** is the set of numbers that will be on your graph. All of your data must fit between the numbers of the scale that you choose.



Axis

Labels

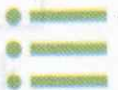
The **labels** will identify what you are showing in your graph. Make sure that you have appropriate labels on the x-axis and on the y-axis.



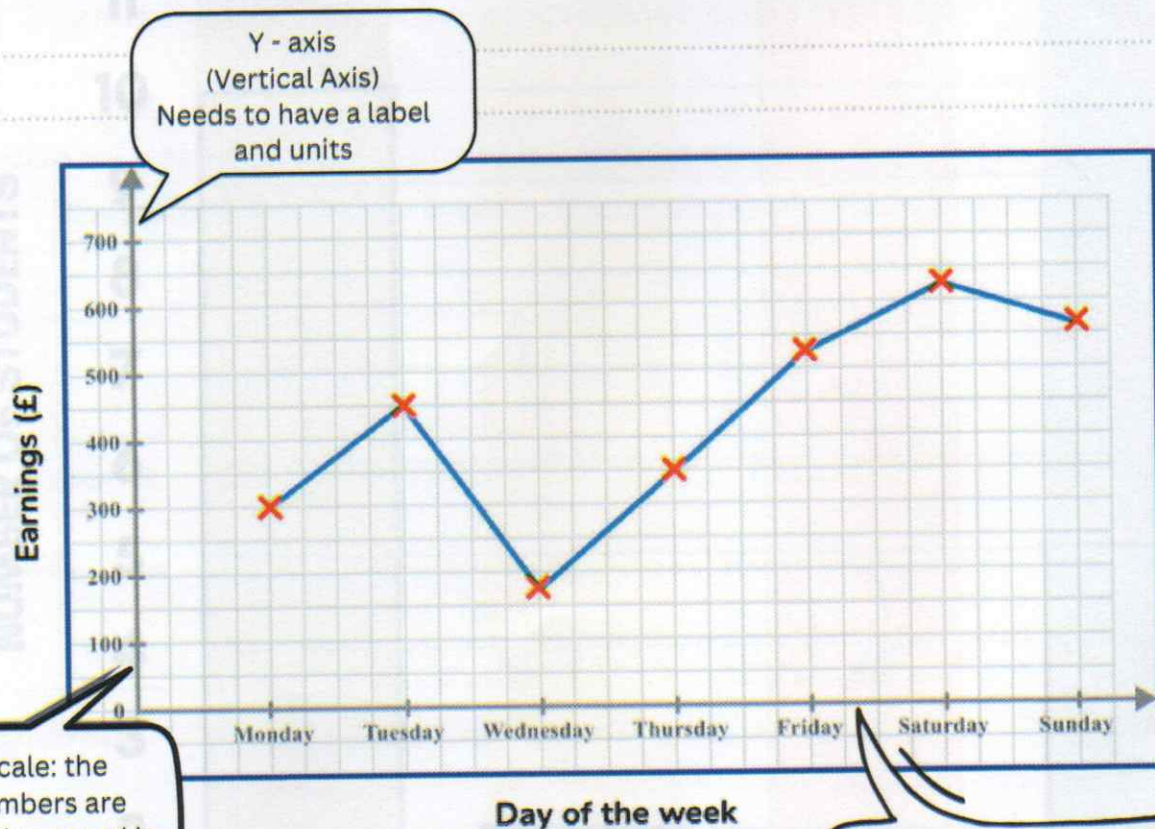
Title

Key

The **key** is used to provide information about your graph, such as what the pictures or bars represent.



Data



Y - axis
(Vertical Axis)
Needs to have a label
and units

Scale: the
numbers are
equally spaced in
a number line.

X - axis
(Horizontal Axis)
Needs to have a label and units

MY CLASS IN GRAPHS

Survey your class to determine the number of students with the following hair colour. Complete the graph, and answer the questions on the following page:

10

Brown

2

Blonde

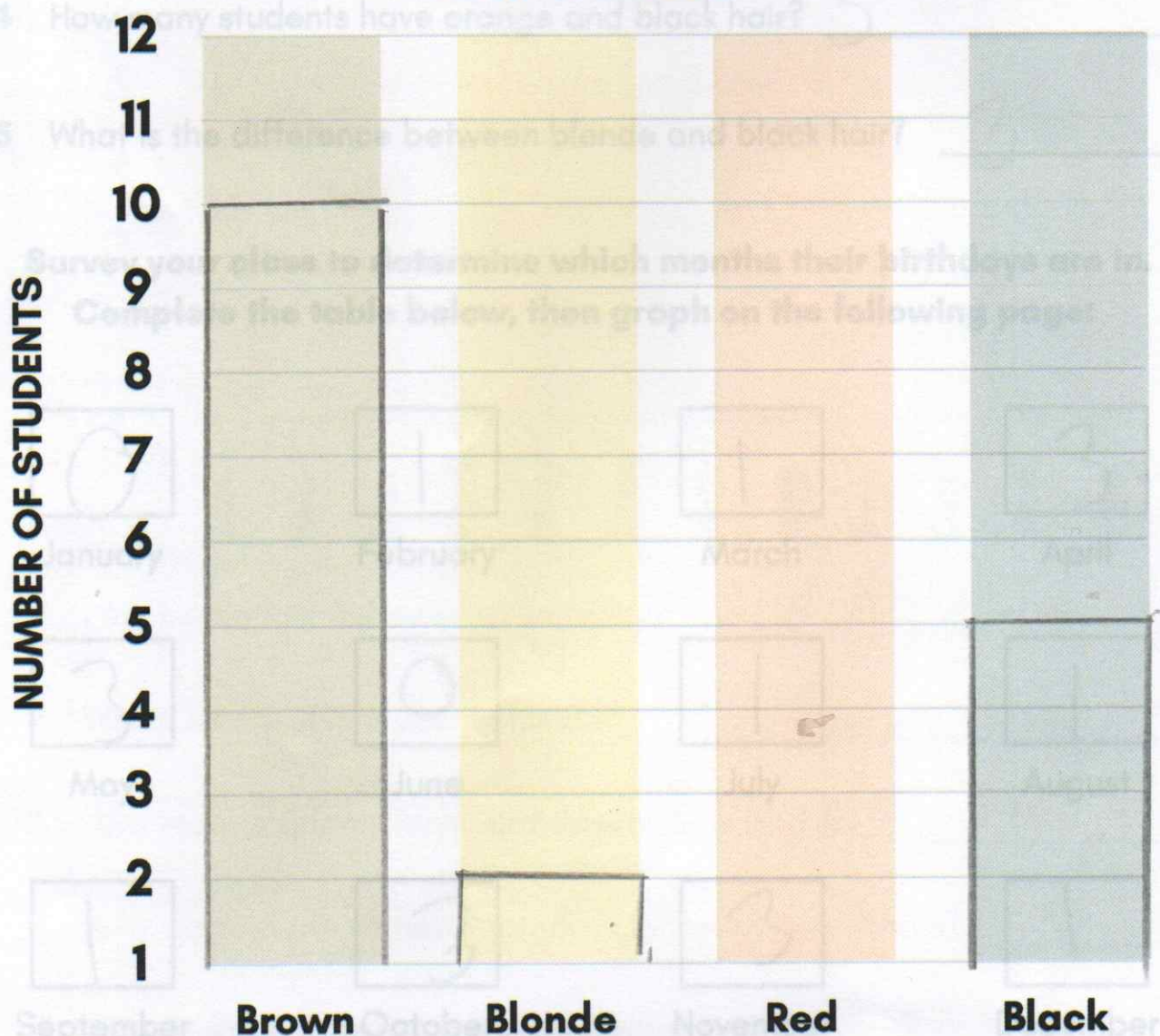
0

Red

5

Black

HAIR COLOUR



MY CLASS IN GRAPHS

Answer the following questions:

- 1 Which hair colour is the most common? Brown
- 2 Which hair colour is the least common? Red
- 3 How many students have blonde and brown hair? 12
- 4 How many students have orange and black hair? 5
- 5 What is the difference between blonde and black hair? 3

Survey your class to determine which months their birthdays are in.
Complete the table below, then graph on the following page:

0

January

1

February

1

March

3

April

3

May

0

June

1

July

1

August

1

September

3

October

3

November

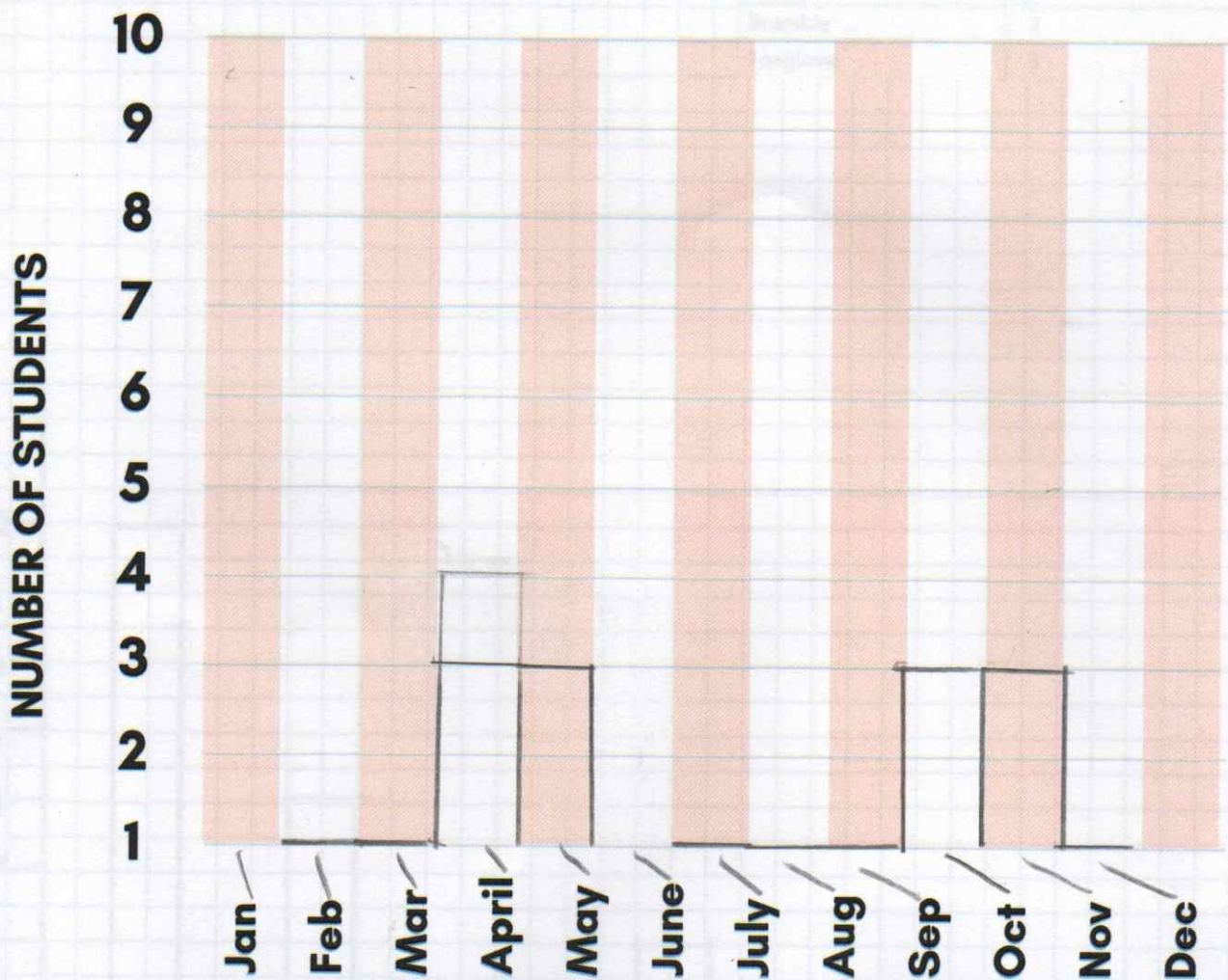
1

December

Graph Number 1

MY CLASS IN GRAPHS

BIRTHDAY MONTHS



1 Which month is the most common?

April, May, October, November

2 Which month is the least common?

January, June

3 How many students have birthdays in June and July?

1

4 How many students have birthdays in the winter months?

4

5 What is the difference between September and May?

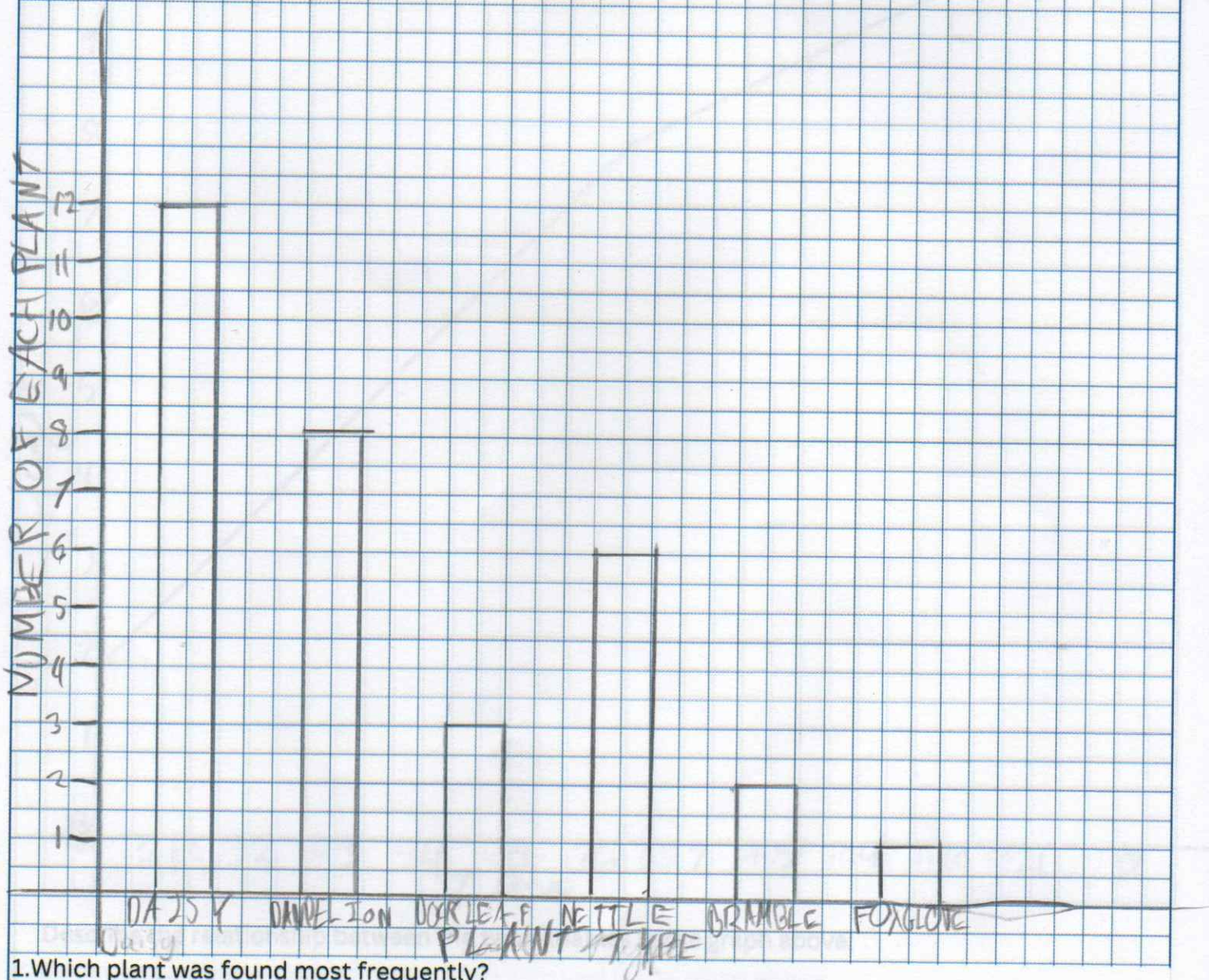
2

Graph Number 1

An ecologist was asked to count the number of each plant that she found in an area of the school garden. The table below shows her results.

Draw an appropriate graph to represent the data.

Plant	Number of plants
Daisy	12
Dandelion	8
Dock Leaf	3
Nettle	6
Bramble	2
Foxglove	1



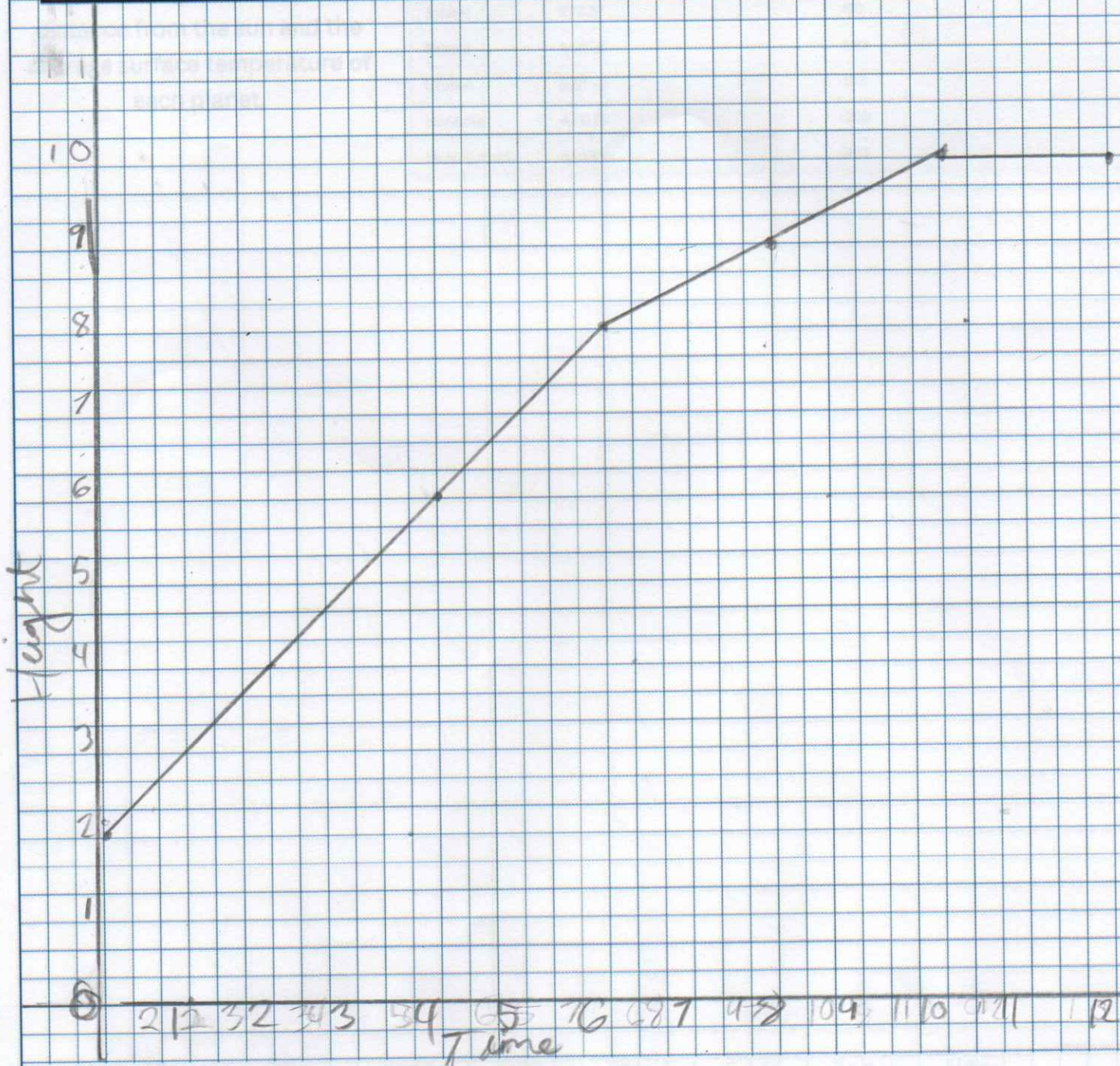
1. Which plant was found most frequently?

2. Which was the scarcest plant in the ecosystem?

Graph Number 2

In this experiment, Tom measured the height of a plant at different time intervals to study its growth. The data he collected is represented in the table below. Construct a graph to best represent Tom's findings.

Time (days)	0	2	4	6	8	10	12
Height (cm)	2	4	6	8	9	10	10



Describe the relationship between the two variables in the graph above:

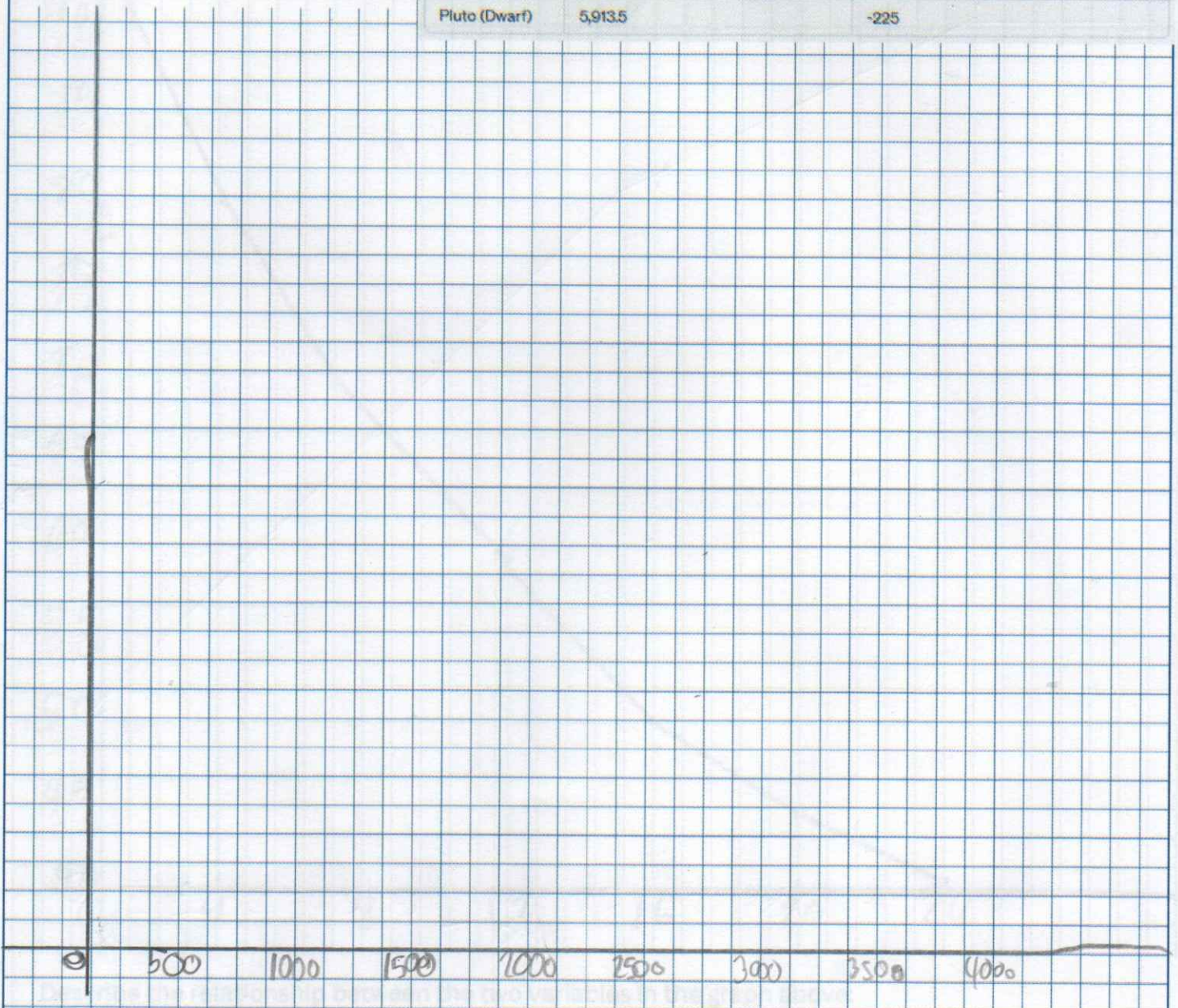
Graph Number 3

To the right is a table of data representing the average surface temperature of the planets and their approximate distance from the sun.

Construct a graph to show the relationship between the distance from the sun and the average surface temperature of each planet.

Table: Surface Temperatures and Distances from the Sun of Solar System Planets

Planet	Distance from Sun (Million km)	Average Surface Temperature (°C)
Mercury	57.9	427
Venus	108.2	467
Earth	149.6	15
Mars	227.9	-65
Jupiter	778.3	-110
Saturn	1,427.0	-140
Uranus	2,871.0	-195
Neptune	4,497.1	-200
Pluto (Dwarf)	5,913.5	-225



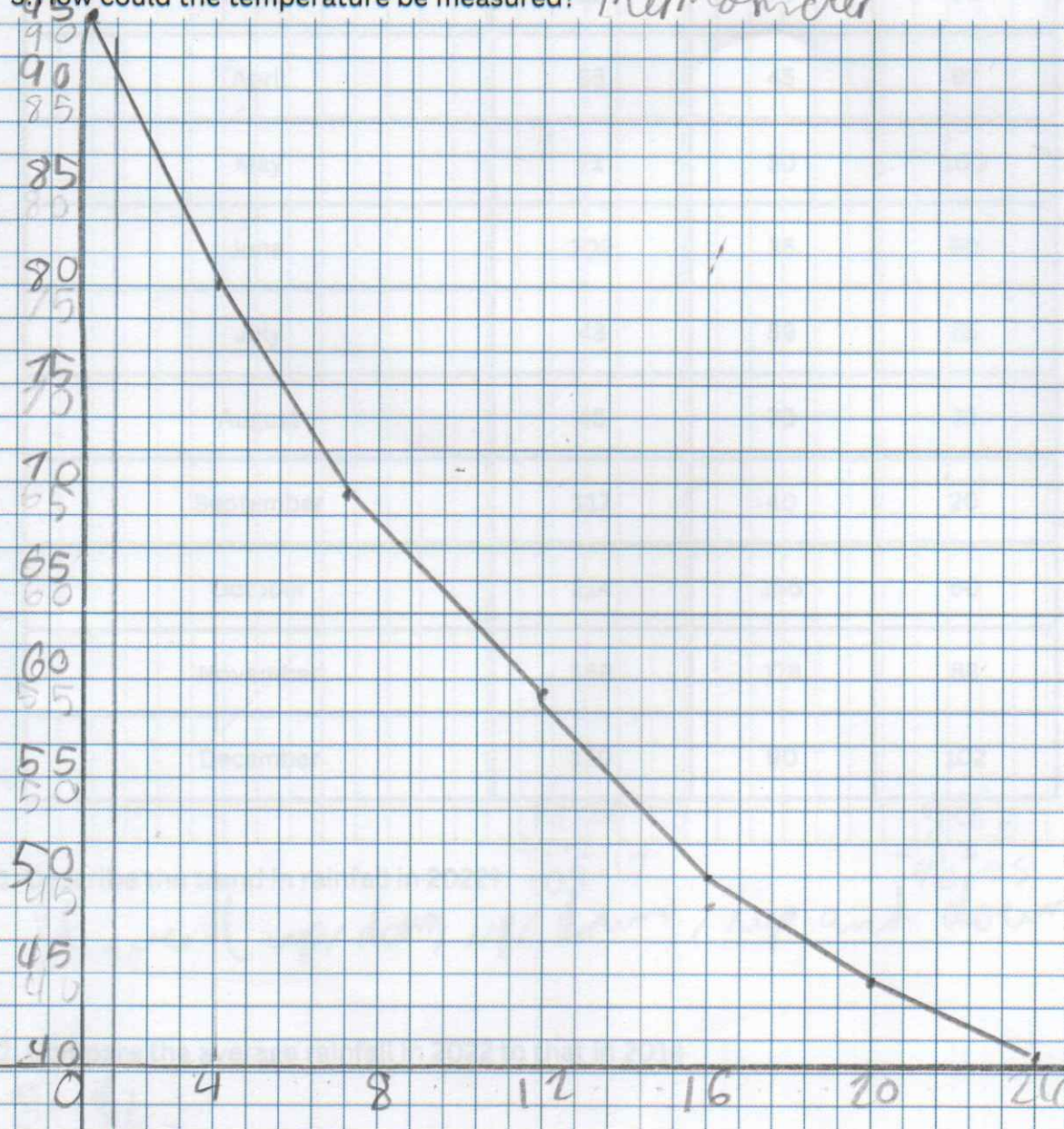
Describe the relationship between the two variables in the graph above:

Graph Number 4

Lucy carried out an investigation to see how the temperature of a jug of hot water changes over time. She measured the temperature every four minutes. She got the results in the table below. Plot a graph to represent the data that Lucy collected during her experiment.

Temperature ($^{\circ}\text{C}$)	95	80	68	58	50	44	40
Time (mins)	0	4	8	12	16	20	24

1. Identify the independent variable: *Time*
2. Identify the dependent variable: *Temp*
3. How could the temperature be measured? *Thermometer*



Describe the relationship between the two variables in the graph above:

Graph Number 5

The table below represents the average rainfall in each month in 2022, 2018 and 2014. Represent this data clearly on a graph and answer the questions that follow.

Month	Average monthly rainfall (mm)		
	2022	2018	2014
January	59	100	89
February	157	71	90
March	65	75	60
April	65	45	97
May	71	30	100
June	102	35	50
July	43	59	55
August	45	70	32
September	117	40	20
October	214	156	60
November	168	178	88
December	120	90	102

1. Describe the trend in rainfall in 2022?

it went up, down, up, down, up and down

2. Compare the average rainfall in 2022 to that in 2014

31.92

3. What month had the highest average rainfall over the course of the three year data?

November

4. In 2018, which month had the lowest rainfall?

September

5. How does representing the data on a graph benefit the audience reading the data?

easier to compare

Makes it

Graph Number 5

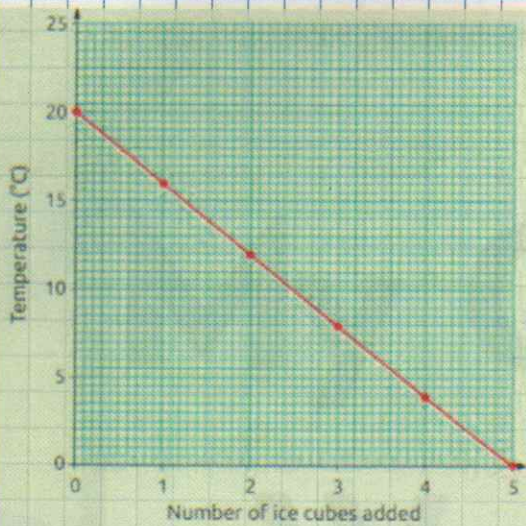
2022



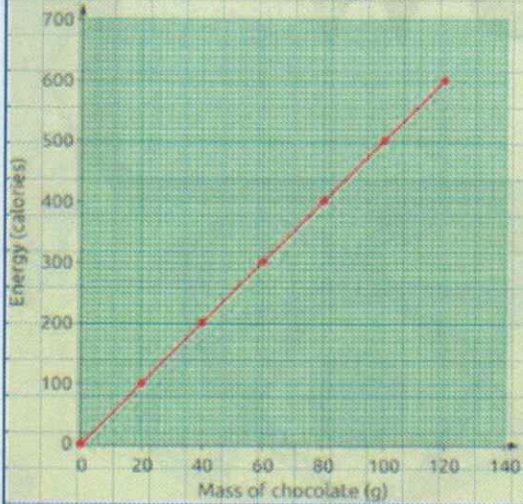
Intrepreting Graphs

Describe the relationship between the data in the graphs below.

1. The temperature of a drink compared to the number of ice cubes added.



2. The amount of energy in a chocolate bar compared with its mass.



3. The mass of a fruit compared with the volume of water received by an apple tree daily.

